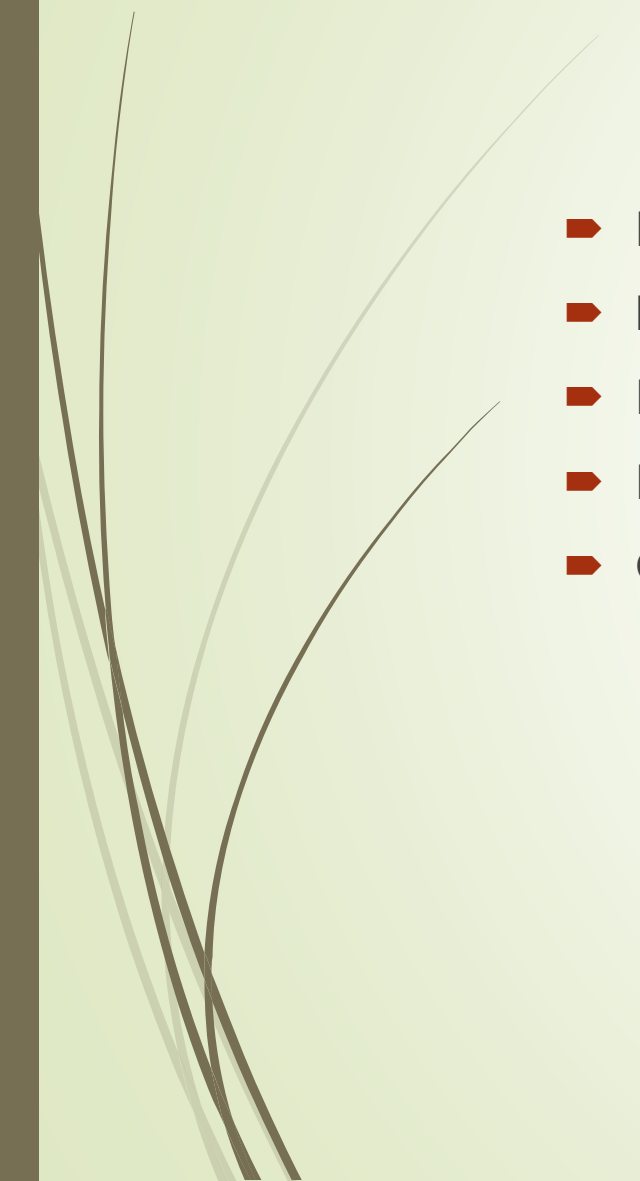


Legal Issues in Computer Security





Outline



- Protecting programs and data: copyrights, patents, trade secrets
- Information and the Law
- Rights of Employees and Employers
- Redress for Software Failures
- Computer Crime

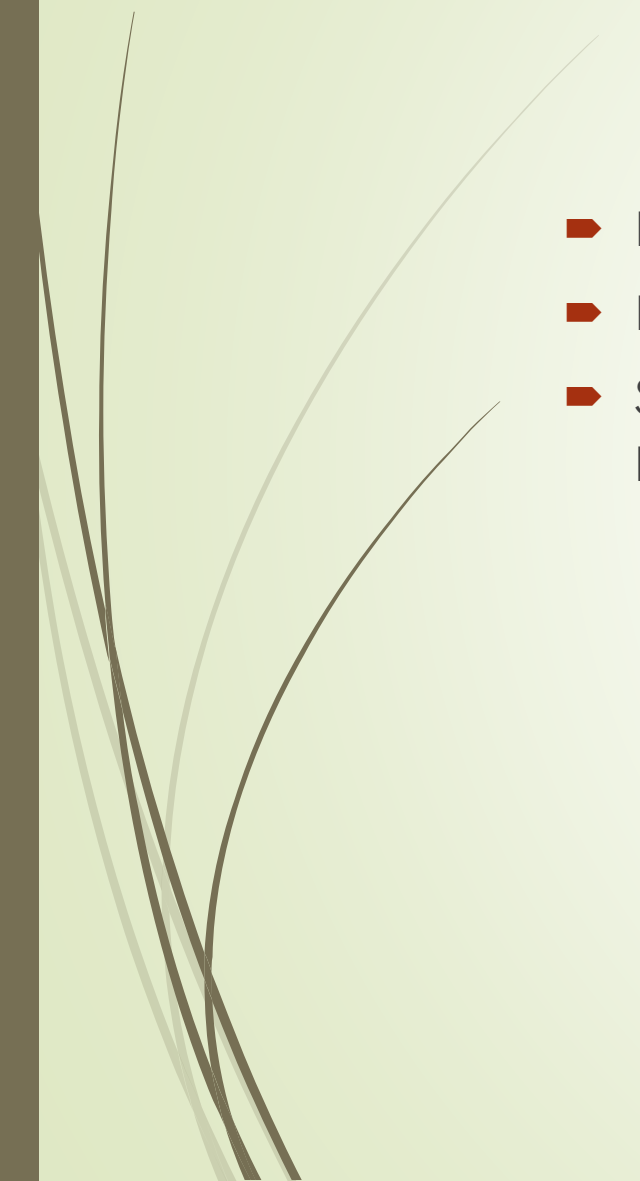


Protecting Programs and Data: Copyrights

- Copyright protects expression of a creative work and promotes exchange of ideas.
- Copyright protects artifacts—expressions of ideas—not the ideas themselves.
- Fair use allows copies for scholarship and research.
- An author or artist profits from the first sale of an object.
- The idea embodied in a copyrighted work must be made public.



Protecting Programs and Data: Patents

- ▶ Patents protect inventions, tangible objects, not their design or idea.
 - ▶ Patent holders must act against all infringers.
 - ▶ Software can be patented, and the courts increasingly recognize the patentability of a novel technique, that is, an algorithm.
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Protecting Programs and Data: Trade Secrets

- A trade secret is unlike a patent or copyright in that it must be kept a secret
- A trade secret is a secret valuable to a business owner. A trade secret is information that gives one company a competitive edge over others.
- If someone obtains a trade secret improperly and profits from it, the owner can recover profits, damages, lost revenues, and legal costs
- Another way trade secret protection can vanish is by reverse engineering
- In reverse engineering, one studies a finished object to determine how it is manufactured or how it works.



Protecting Programs and Data: Trade Secrets

- Trade secret protection applies very well to computer software. The underlying algorithm of a computer program is novel, but its novelty depends on nobody else's knowing it.
- Trade secret protection allows distribution of the result of a secret (the executable program) while keeping the program design hidden.
- The difficulty with computer programs is that reverse engineering works. Decompiler and disassembler programs can produce a source version of an executable program.
- Trade secret protection is of no help when someone infers a program's design by studying its output or, worse yet, decoding the object code. Both are legitimate (that is, legal) activities, and both cause trade secret protection to disappear.



Protecting Programs and Data: Special Cases

- Computer Source or Object Code
 - Computer source or object code can also be protected as intellectual property under copyright. However, that protection applies only to the reproductions of the code, not to a variation of the concept.
 - Reselling a copy of a piece of software is a violation, but rewriting a piece of code, that is, reimplementing the same algorithm, is within the limits of copyright law.
- Web Content
 - You may purchase or use with permission a piece of web art, a widget (such as an applet that shows a spinning globe), or some music. Copyright protects your original works.
- Domain Names and URLs
 - Domain names, URLs, company names, product names, and commercial symbols are protected by a trademark, which gives exclusive rights of use to the registered owner of such identifying marks.



Information and the Law: Information as an Object

- Information Is Not Depletable
 - Information has value unrelated to whatever medium contains it.
- Information Can Be Replicated
- Information Has a Minimal Marginal Cost
 - The marginal cost of an item is the cost to produce another one after having produced some already
 - The marginal cost of information is often minuscule.
- The Value of Information Is Often Time Dependent
- Information Is Often Transferred Intangibly



Information and the Law: Legal Issues Relating to Information

- Laws do not, and cannot, control all cyber threats. Let us look at several examples of situations in which information needs are about to place significant demands on the legal system.
 - Information Commerce
 - Electronic Publishing
 - Protecting Data in a Database
 - Electronic Commerce



Information and the Law: The Legal System

- Criminal and Civil Law
 - Criminal law involves a wrongful action against society.
 - Civil law involves harm to an individual or a corporation.
- Tort Law
 - A tort is harm not occurring from violation of a statute or from breach of a contract but instead from being counter to the accumulated body of precedents. Tort law is unwritten but evolves through court decisions that become precedents for cases that follow.
- Contract Law
 - A contract is an agreement between two parties. A contract must involve three things:
 - an offer
 - an acceptance
 - a consideration
 - Computer contracts typically involve the development and use of software and computerized data. There are rules about who has the right to contract for software—employers or employees—and what are reasonable expectations of software's quality.



Rights of Employees and Employers: Ownership of Products

- An employment contract clarifies for both parties an employee's rights to computer products.
 - Ownership of a Patent: Under patent law, it is important to know who files the patent application. If an employee lets an employer patent an invention, the employer is deemed to own the patent and therefore the rights to the invention.
 - Ownership of a Copyright: Owning a copyright is like owning a patent. The author (programmer) is the presumed owner of the work, and the owner has all rights to an object. However, a special situation known as work for hire applies to many copyrights for developing software or other products.
 - In a work for hire, the employer is the creator and owner of an employee's work product.
 - An alternative to a work for hire arrangement is licensed software. In this situation, the programmer develops and retains full ownership of the software.
 - A company owns the trade secrets of its business-confidential data. As soon as a secret is developed, the company becomes the owner.



Rights of Employees and Employers: Employment Contracts

- Typically, an employment contract specifies that the employee be hired to work as a programmer exclusively for the benefit of the company.
- The company claims all rights to any programs developed, including all copyright rights and the right to market.
- More restrictive contracts (from the employee's perspective) assign to the employer rights to all inventions (patents) and all creative works (copyrights), not just those that follow directly from one's job.
- An agreement not to compete is sometimes included in a contract. The employer states that simply having worked for one employer will make the employee very valuable to a competitor. The employee agrees not to compete by working in the same field for a set period after termination.



Redress for Software Failures

- Selling Correct Software: You may have following reactions if you find something wrong with software;
 - I Want a Refund: Software is supposed to be returnable for a refund in a reasonable time.
 - I Want It to Be Good: Product must be usable for its intended purpose; software that does not work is clearly not usable.
- Reporting Software Flaws: Disclosing vulnerabilities encourages vendors to develop and disseminate patches but patching under time pressure is counter to fixing flaws completely.

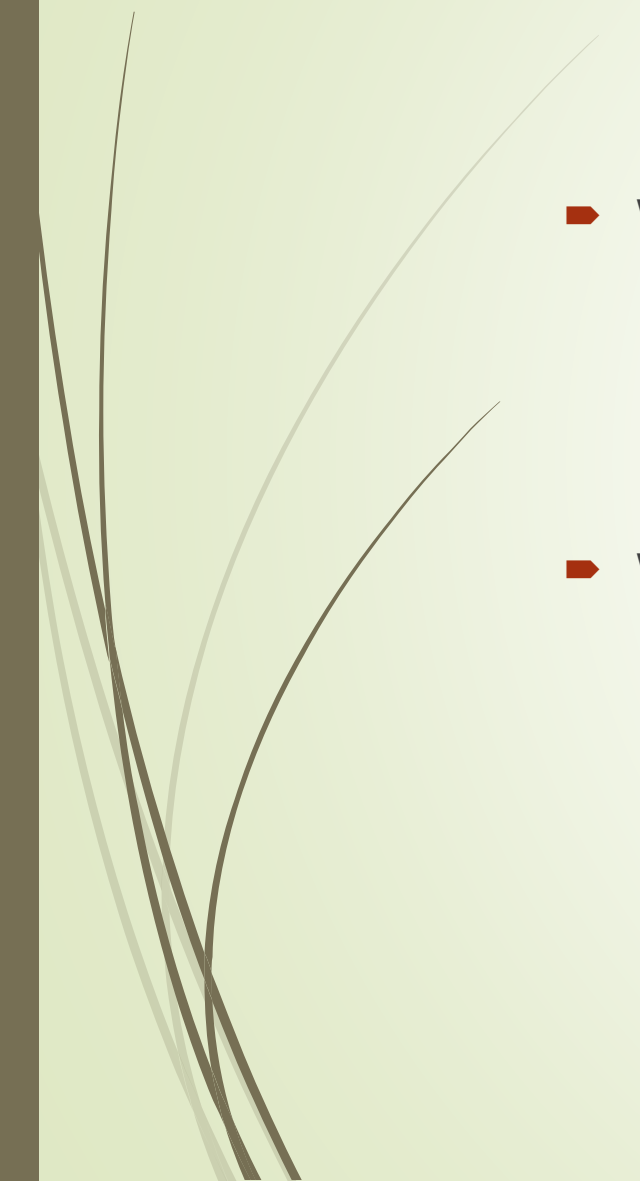


Computer Crime

- Crimes involving computers are an area of the law that is even less clear than the other areas
- Rules of Property
 - The legal system has explicit rules about what constitutes property.
 - To a computer professional, taking a copy of a software package without permission is clear-cut theft
 - We would generally agree that unauthorized access to a computing system is a crime
- Rules of Evidence
 - Computer printouts have been used as evidence in many successful prosecutions.
 - A computer printout showing activity around a time period of interest is solid evidence, especially if a system administrator or other person in charge can testify that the system generates this log data continually, and this printout is an accurate depiction of the contents of the log file.
 - Forensic analysis is a field in which computer security experts examine artifacts such as disk drives, log files, program code, even volatile memory, to discern facts about data contained.
- Value of Data
 - A person was found guilty of having stolen a substantial amount of data from a computer data bank.
 - Digital data, like many other things of value, is worth what a willing buyer would pay for it.



Computer Crime



- Why Computer Crime Is Hard to Define
 - Creating and changing laws are slow processes, intended to involve substantial thought about the effects of proposed changes. This deliberate process is very much out of pace with a technology that is progressing as fast as computing.
 - A particular computer can be the subject, object, or medium of a crime. A computer can be attacked (attempted unauthorized access), used to attack (impersonating a legitimate node on a network), and used to commit crime (Trojan horse or fake login).
- Why Computer Crime Is Hard to Prosecute
 - Lack of understanding
 - Lack of physical evidence
 - Lack of political impact
 - Complexity of case
 - Age of defendant



Computer Crime

- ▶ Why Computer Criminals Are Hard to Catch

- ▶ As if computer crime laws and prosecution were not enough, it is also difficult for law enforcement agencies to catch computer criminals. There are two major reasons for this.
 - ▶ First, computer crime is a multinational activity that must usually be pursued on a national or local level. There are no international laws on computer crime
 - ▶ Complexity is an even more significant factor than country of origin. Networked attacks are hard to trace and investigate because they can involve so many steps. Each step along the way makes the investigator complete more legal steps.

- ▶ What Computer Crime Does Not Address

- ▶ Legislators cannot define precisely what a computer is because computer technology is used in many other devices, such as robots, calculators, watches, automobiles, microwave ovens, and medical instruments.



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